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Internship Organization: CyArt
Internship Domain: SOC (Cyber Security)
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TITTLE

SOC Practical: Log Analysis and Security Monitoring

OBJECTIVE

The objective of this task is to understand Security Operations Center (SOC) fundamentals and perform log analysis to detect failed login attempts and potential brute-force attacks. Additionally, browser history analysis is conducted to identify any suspicious web activity.

SOC FUNDAMENTALS

A Security Operations Center (SOC) is a centralized team responsible for monitoring, detecting, and responding to cybersecurity threats in real time.
The main purpose of a SOC is to ensure proactive threat detection, incident response, and continuous monitoring of systems.
SOC roles include:
- Tier 1 Analyst: Monitors alerts and performs initial analysis
- Tier 2 Analyst: Investigates incidents in detail
- Tier 3 Analyst: Performs threat hunting and advanced analysis

Key functions of a SOC include log analysis, alert triage, and threat intelligence integration.

SECURITY MONITORING BASICS

Security monitoring is the process of continuously observing systems and networks to detect suspicious activities, unauthorized access, and policy violations.

The main objectives of security monitoring are:

- Detect anomalies in system behavior
- Identify unauthorized access attempts
- Ensure system security and compliance

Common tools used in security monitoring include:

- :contentReference[oaicite:0]{index=0} (SIEM tool for log analysis)
- :contentReference[oaicite:1]{index=1} (open-source monitoring tool)
- :contentReference[oaicite:2]{index=2} (network traffic analysis)

Key metrics include false positives, false negatives, and Mean Time to Detect (MTTD).

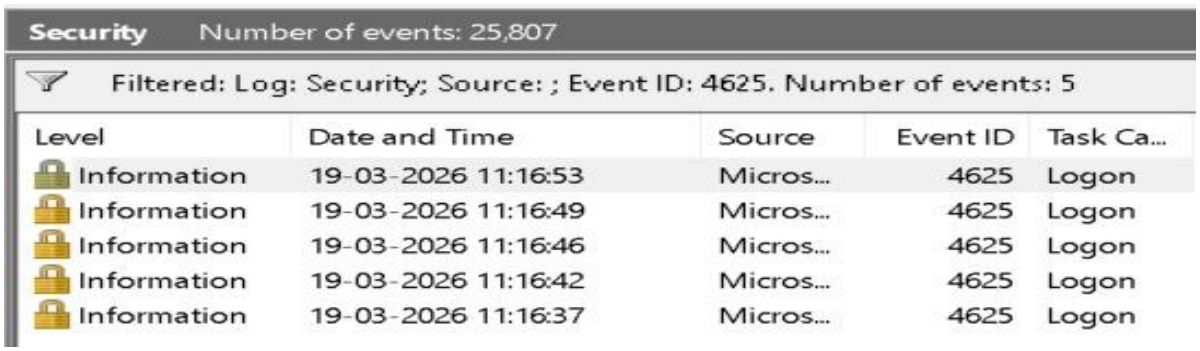
LOG MANAGEMENT FUNDAMENTALS

Log management involves the collection, storage, and analysis of log data generated by systems and applications.
The log lifecycle includes:
- Collection: Gathering logs from different sources
- Normalization: Converting logs into a standard format
- Storage: Saving logs securely
- Analysis: Examining logs to detect suspicious activities

Common types of logs include Windows Event Logs, Syslog, and HTTP server logs.
In this task, Windows Event Logs were used to analyze security events and detect failed login attempts.

LOG ANALYSIS PRACTICE

In this task, log analysis was performed using Windows Event Viewer to detect failed login attempts.
Multiple failed login attempts were generated by entering incorrect passwords on the system. These logs were then analyzed to identify suspicious activity.
Event ID 4625 was used to filter failed login attempts from the Security logs.



Security Number of events: 25,807

Filtered: Log: Security; Source: ; Event ID: 4625. Number of events: 5

Level	Date and Time	Source	Event ID	Task Ca...
Information	19-03-2026 11:16:53	Micros...	4625	Logon
Information	19-03-2026 11:16:49	Micros...	4625	Logon
Information	19-03-2026 11:16:46	Micros...	4625	Logon
Information	19-03-2026 11:16:42	Micros...	4625	Logon
Information	19-03-2026 11:16:37	Micros...	4625	Logon

DETAILED LOG ANALYSIS

A detailed analysis of a failed login event showed that the login attempt failed due to an incorrect password. The target account was 'Dell' and the source IP was 127.0.0.1, indicating a local login attempt.

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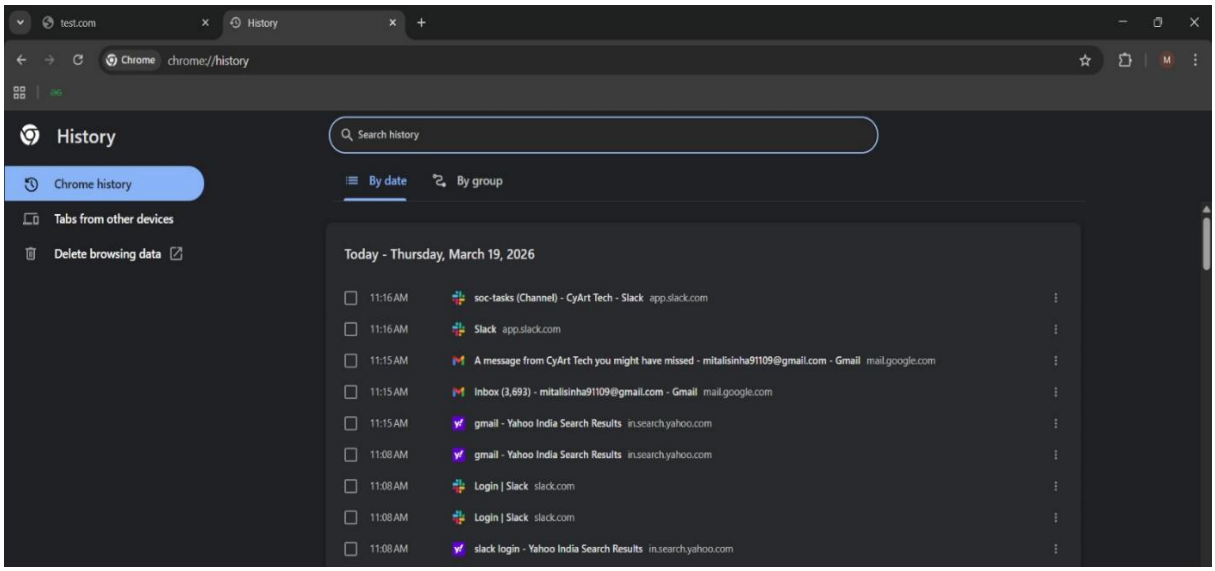
Information	19-03-2026 11:16:37	Micros...	4625	Logon

BROWSER HISTORY ANALYSIS

Browser history was analyzed using Google Chrome to identify suspicious URLs. The search for test.com returned no results, indicating no suspicious activity.



LOG COLLECTION USING ELASTIC STACK

Logs were collected and monitored using Elastic Stack. Filebeat was used to ship logs, and Elasticsearch was used for storing and indexing the logs.

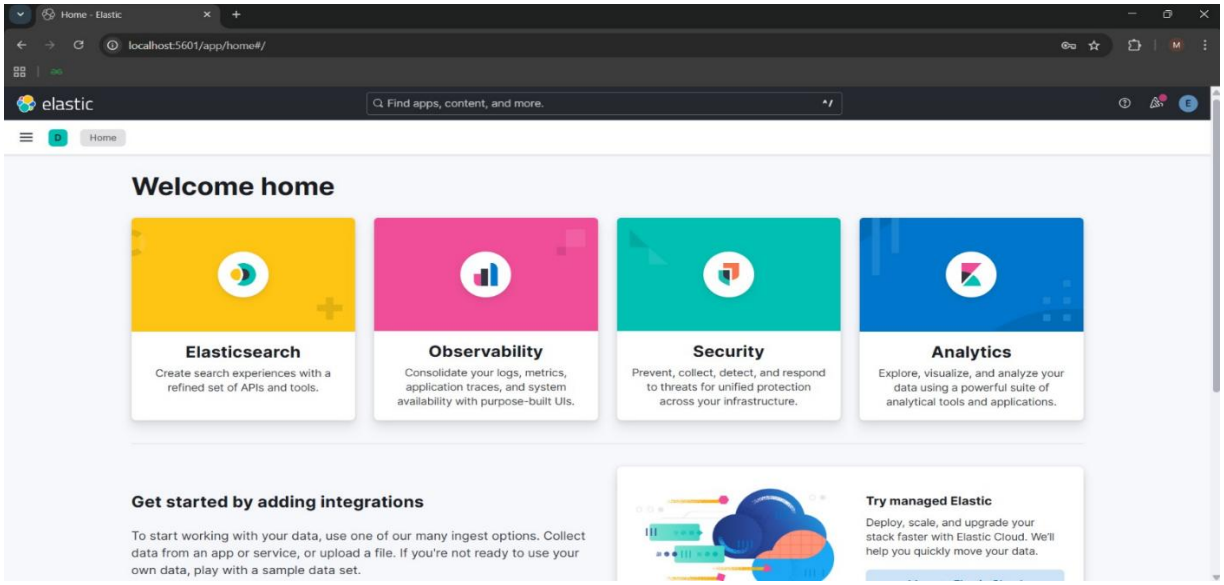
```
● filebeat.service - Filebeat sends log files to Logstash or directly to Elasticsearch.
     Loaded: loaded (/usr/lib/systemd/system/filebeat.service; disabled; preset: enabled)
     Active: active (running) since Thu 2026-03-19 07:37:04 UTC; 19s ago
       Docs: https://www.elastic.co/beats/filebeat
   Main PID: 1265 (filebeat)
      Tasks: 13 (limit: 4504)
     Memory: 159.6M (peak: 160.5M)
        CPU: 490ms
     CGroup: /system.slice/filebeat.service
             └─1265 /usr/share/filebeat/bin/filebeat --environment systemd -c /etc/filebeat/filebeat.yml --path.home /usr/share/filebeat --path.config /etc>

Mar 19 07:37:05 LAPTOP-J459LKIC filebeat[1265]: {"log.level":"info","@timestamp":"2026-03-19T07:37:05.473Z","log.logger":"monitoring","log.origin":{"functi>
Mar 19 07:37:05 LAPTOP-J459LKIC filebeat[1265]: {"log.level":"info","@timestamp":"2026-03-19T07:37:05.473Z","log.origin":{"function":"github.com/elastic/be>
Mar 19 07:37:05 LAPTOP-J459LKIC filebeat[1265]: {"log.level":"info","@timestamp":"2026-03-19T07:37:05.483Z","log.logger":"filebeat","log.origin":{"function>
Mar 19 07:37:05 LAPTOP-J459LKIC filebeat[1265]: {"log.level":"info","@timestamp":"2026-03-19T07:37:05.488Z","log.logger":"registrar","log.origin":{"functio>
Mar 19 07:37:05 LAPTOP-J459LKIC filebeat[1265]: {"log.level":"info","@timestamp":"2026-03-19T07:37:05.489Z","log.logger":"crawler","log.origin":{"function">
Mar 19 07:37:05 LAPTOP-J459LKIC filebeat[1265]: {"log.level":"info","@timestamp":"2026-03-19T07:37:05.490Z","log.logger":"crawler","log.origin":{"function">
Mar 19 07:37:05 LAPTOP-J459LKIC filebeat[1265]: {"log.level":"info","@timestamp":"2026-03-19T07:37:05.490Z","log.logger":"crawler","log.origin":{"function">
Mar 19 07:37:05 LAPTOP-J459LKIC filebeat[1265]: {"log.level":"info","@timestamp":"2026-03-19T07:37:05.491Z","log.logger":"crawler.module.reloader","log.ori>
Mar 19 07:37:05 LAPTOP-J459LKIC filebeat[1265]: {"log.level":"info","@timestamp":"2026-03-19T07:37:05.491Z","log.logger":"crawler.module.reloader","log.ori>
Mar 19 07:37:08 LAPTOP-J459LKIC filebeat[1265]: {"log.level":"info","@timestamp":"2026-03-19T07:37:08.444Z","log.logger":"processors.add_cloud_metadata","l>
~
~
~
~
~
```

```
mitali1804@LAPTOP-J459LKIC:~$ sudo service filebeat restart
mitali1804@LAPTOP-J459LKIC:~$ curl -k -u elastic https://localhost:9200/_cat/indices?v
Enter host password for user 'elastic':
health status index                                                    uuid                   pri rep docs.count docs.deleted store.size pri.store
.size dataset.size
green  open   .internal.alerts-transform.health.alerts-default-000001  i_pKyVdMTde1ItSMl3PYfw   1   0          0            0       249b
 249b         249b
green  open   .internal.alerts-observability.logs.alerts-default-000001 v7OxnA3iQaasSRBMtnsK1w   1   0          0            0       249b
 249b         249b
green  open   .internal.alerts-observability.uptime.alerts-default-000001 eyGXpi7tQee5kRtaqKegMw   1   0          0            0       249b
 249b         249b
green  open   .internal.alerts-ml.anomaly-detection.alerts-default-000001 FnMu6Rx9QC6nEsCRS6xGTQ   1   0          0            0       249b
 249b         249b
green  open   .internal.alerts-observability.slo.alerts-default-000001 D2RmsKXbTeuKhNtlaofwWA   1   0          0            0       249b
 249b         249b
green  open   .internal.alerts-default.alerts-default-000001           R9rTThhTTSqtWbU5HT-Xrw   1   0          0            0       249b
 249b         249b
green  open   .internal.alerts-streams.alerts-default-000001           895d7lwEQiuKOojtxvtyDQ   1   0          0            0       249b
 249b         249b
green  open   .internal.alerts-observability.apm.alerts-default-000001 1vTRgCnQROaY4zyi8idctg   1   0          0            0       249b
 249b         249b
green  open   .internal.alerts-security.attack.discovery.alerts-default-000001 X4UaiEZVT4m_qhQdcqm6yA   1   0          0            0       249b
 249b         249b
green  open   .internal.alerts-observability.metrics.alerts-default-000001 d3jAZk6RQAi02N2wUEBBcg   1   0          0            0       249b
 249b         249b
green  open   .internal.alerts-ml.anomaly-detection-health.alerts-default-000001 GUTF8NPqQkGaeWRe_fpnAg   1   0          0            0       249b
 249b         249b
green  open   .internal.alerts-observability.threshold.alerts-default-000001 70-RjtOYRV-JIS__IZItTA   1   0          0            0       249b
 249b         249b
green  open   .internal.alerts-security.alerts-default-000001          d2PW4pNTTM6YMODMmeS16g   1   0          0            0       249b
 249b         249b
green  open   .internal.alerts-dataset.quality.alerts-default-000001  UyRevUKeTQKWtU3OM_6O7A   1   0          0            0       249b
 249b         249b
yellow open   .ds-filebeat-8.19.12-2026.03.19-000001                   mwtul2L8SSyhXPR-oZvMHg   1   1          0            0       249b
 249b         249b
green  open   .internal.alerts-stack.alerts-default-000001             RPRrbQgsROmkrPhj_EyUvA   1   0          0            0       249b
 249b         249b
```

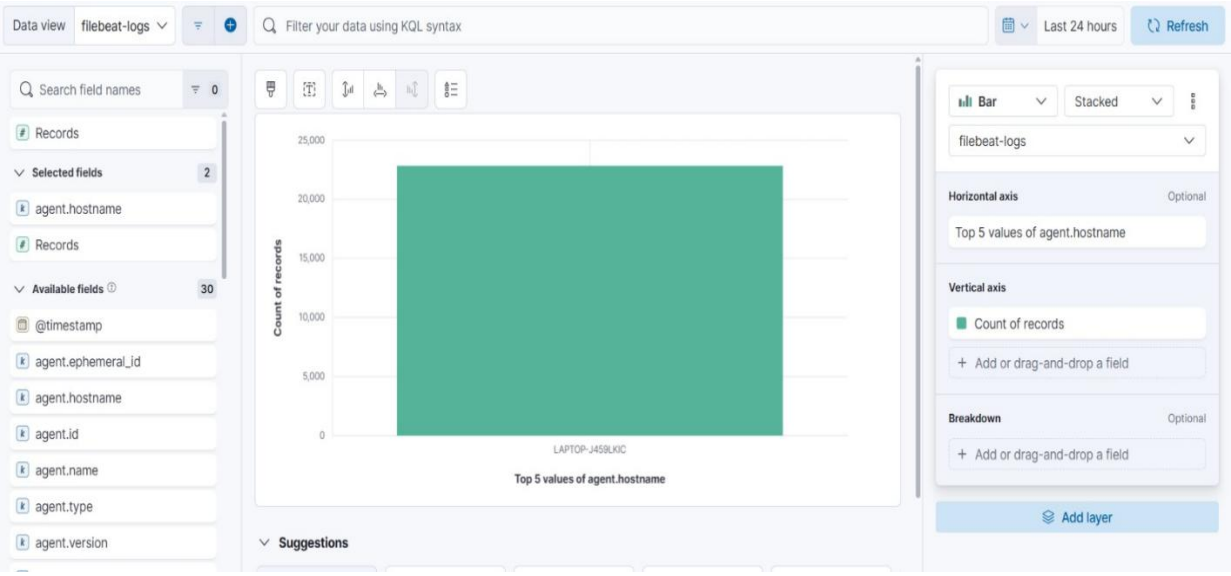
SIEM DASHBOARD (ELASTIC UI)

The logs were visualized using the Elastic SIEM dashboard. Various dashboards and tools were used to analyze log data.



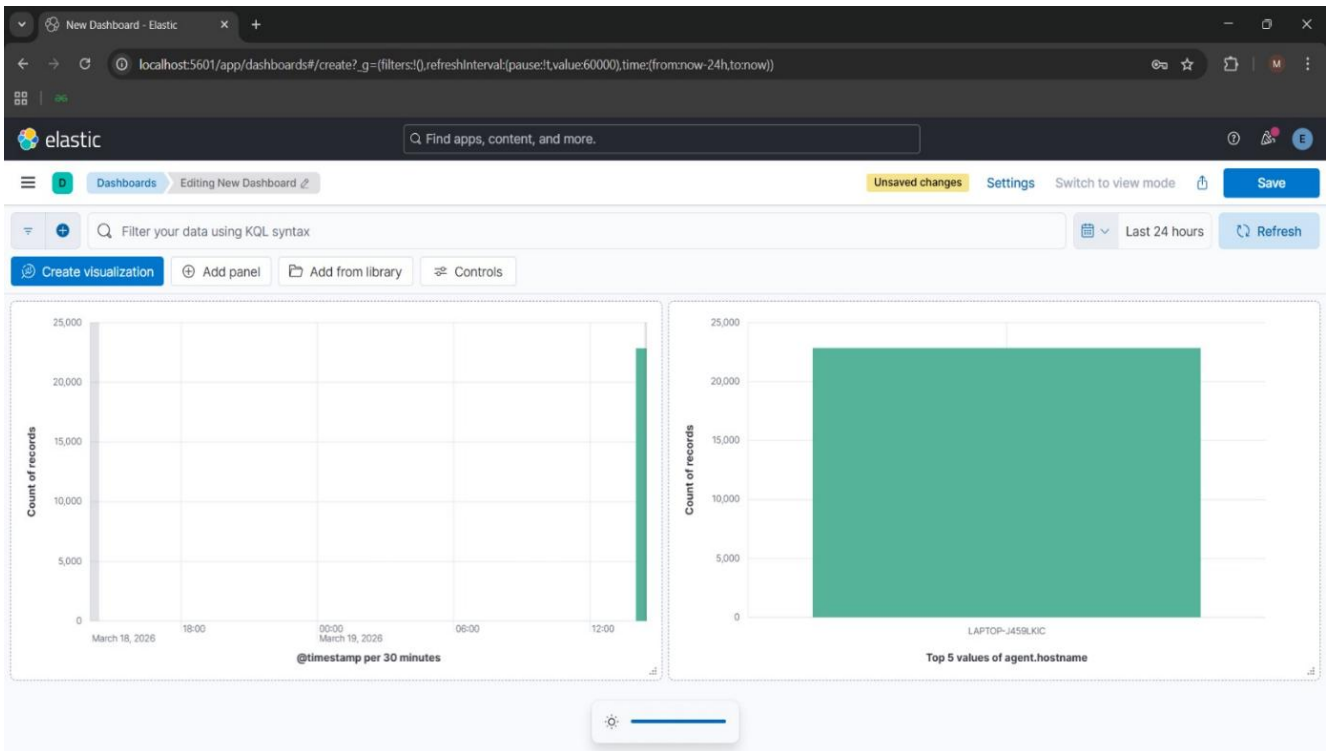
LOG VISUALIZATION

Log data was visualized using charts to analyze patterns such as frequency of logs and system activity over time.



DASHBOARD CREATION

Custom dashboards were created to monitor system activity and visualize important log metrics.



CONCLUSION

This project demonstrated the practical implementation of SOC concepts including log analysis, security monitoring, and alert detection. Failed login attempts were successfully identified using Windows Event Viewer, indicating potential brute-force attacks. Additionally, browser history analysis confirmed no suspicious activity. The use of Elastic Stack further enhanced log monitoring and visualization capabilities, allowing better understanding of system behavior through dashboards and graphs. This task highlights the importance of continuous monitoring and proactive threat detection in cybersecurity.

TOOLS USED

Tools Used:
- Windows Event Viewer
- Google Chrome
- Elastic Stack (Elasticsearch, Kibana)
- Filebeat